

Element H: Prototype testing and data collection plan

Simplicity of latching/opening - 0.15

Latching ability - 0.15

X-ray Compatibility - 0.25

Range of operator motion - 0.2 1/13

Durability - 0.15

Reliability - 0.1

Side Door

1. Ease/Simplicity of Opening and Closing

Quantitative Data Measured: Opening/Closing Time

Qualitative Data Measured: Ratings of opening and closing experience

We chose the testing method of opening and closing times to test the usability of the window during operation and the ability to make quick decisions in emergency situations. Part of measuring “ease” of use is being able to make quick decisions. Additional to this, team members observed a lack of qualitative or client data that may help in fitting the needs of the zookeepers. To fill this gap in data, a survey will be conducted about the satisfaction of test subjects during their use of the prototype. These subjects will convey their reactions through a one to five rating system about their experience and whether the movement is smooth and satisfactory.

Test Method 1 - Opening/Closing Time(s)

E.g. open/close half way, open/close up one notch, open/close fully

1. Place the testing instrument (sliding door) upright on a table that rests at hip height.
2. Place door in closed orientation as shown in Figure 1
3. Recruit an individual (“Test Subject”) to test prototype
4. Have Test Subject select a distance away from testing instrument deemed comfortable to reach the side door
5. Have Test Subject place arms straight down to their side

6. Instruct Test Subject, upon hearing the word “Start,” to open the side door and secure the side door in an open position by the fastest, most apparent means available to them. Once left in an open position, the Test Subject is to place their arms back to their sides.
7. Instruct Test Subject that, immediately following putting their arms to their sides, they must then close the side door and secure the door in a closed position by the fastest, most apparent means available to them. They must then place their arms by their sides.
8. Using a stopwatch with a lap feature, say “Start” and begin the stopwatch timer.
9. Once the Test Subject has placed their arms by their side (following the opening of the door), lap the timer for the first time.
10. Once the Test Subject has closed the door and placed their arms by their side, stop the timer.
11. Record the lap time and stop time of the timer.
12. Consult the Test Subject and ask them to:
 - a. Rate the experience on scale of 1 to 5 in the following categories:
 - i. Motor Effort Needed (1-Significant Effort/Motor Strength Needed; 5-Minimal Effort Needed)
 - ii. Overall Rating
 - b. Write approximately 3 sentences of feedback describing the mechanism’s good parts, bad parts, and places for improvement.

2. Latching Ability

“Latching ability” and “Simplicity of latching/opening” refer to the security of the lock. With continued use of this feature during check-ups for dragons other than the resident female dragons, it is of high importance to ensure the safety of the lock on the opening. The lock must remain secure at all times to guarantee the safety of veterinary technicians. If the lock were to fail, any zoo employee around the box would be put in danger of attacks made by enclosed dragons as this would allow for the claws or tail to be exposed. In the prevention of this event, the simplicity of the lock should be prioritized to streamline the process of closing and locking the opening.

Team members decided to test latching ability via testing required motor effort, otherwise how many hands that we have to use to open it, and testing maximum door force, or how much weight can be applied downward on the handle before the door fails and opens.

Test Method 1 - Required motor effort

Eg # of hands

Test Method 2 - Maximum door force

1. Place the side door in an open position, latched with both spring-mechanisms on either side. The spring which has its end piece adjusted to prevent the door from closing will be the "Open Latch;" the spring with its end piece adjusted to allow the door to close will be the "Close Latch." See Figure 2.1 below.

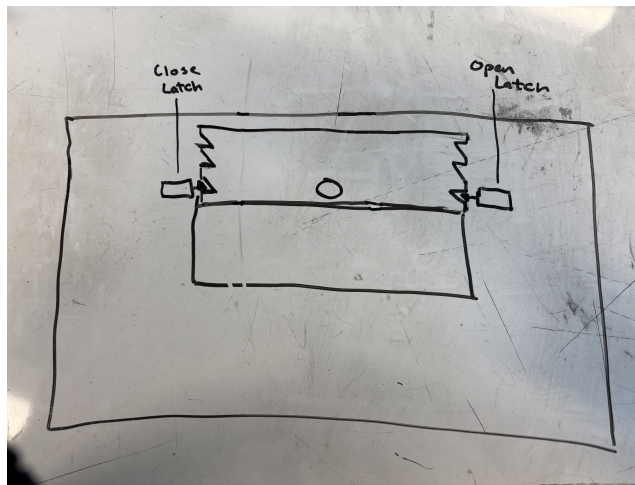


Figure 2.1. Side door assembly.

2. Attach a spring scale to the handle of the door
3. To the end of the spring scale, attach a 10g mass
4. If the side door is still open, check the corresponding box for the given total-mass in Table 2.1 to show the door has passed that test.
5. Repeat steps 3-4 until the side door has been forced shut.
6. Repeat steps 1-5, but secure the door with only the Open Latch. Mark data in the corresponding columns.
7. Repeat steps 1-5, but secure the door with only the Close Latch. Mark data in the corresponding columns.
8. Measure mass of door.

Table 2.1: Latch Mass Tolerance

Mass	Spring Latches Active		
Total Mass (g)	Both	Open	Close
10			
20			
30			
40			
50			
60			
70			

3. X-Ray Compatibility

Expert Engineer 2 from Vanderbilt reviewed our prototype and provided us with images of x-rayed plastics and their respective films. He offered to x-ray our prototype and told us that he would send the film back to give us reference of the effectiveness of the build. After reviewing our conversation and reaching out a couple months after this conversation, Expert Engineer 2 couldn't find an x-ray machine that Brentwood students could use, so instead the X-ray will be provided by the zoo's machinery. In an attempt to achieve clarity in the film, tests of amount and percent area of film covered by x-ray artifacts will be conducted. These tests aim to model the compatibility of the design with x-ray technology and allow zookeepers to image the dragons safely and accurately.

Test Method 1 - Count of X-Ray Artifacts

1. Completely empty interior of box
2. Take box segment with prototype to zoo
3. Place X-ray detector plate against the opposite side of the prototype pane
4. Turn on the x-ray and scan through the prototype pain
5. Analyze the film visible to the tech and count the number of artifacts on the screen

Test Method 2 - Letter grading of film by experts

1. Once the initial X-ray is taken, have the technician download the film onto a disc
2. Send film to 3 experts to letter grade based on a certain rubric of grading criteria
3. Record data in a table and compare evaluations made by experts

4. Range of Operator Motion

Testing range of operator motion is an important aspect of testing the overall effectiveness of the design. A range of motion tests and rating of user experience will be used to determine if this design is well integrated into the entirety of the box. Based on the result of the tests, team

members and stockholders will gain a better understanding of how much area an ultrasound technician can reach while using multiple different wands.

Test Method 1 - Range of Motion Test

1. Place the prototype plate parallel to the table's edge and hold it against the table for the test subject to reach their hand through.
2. Use 3-D printed wands and extension cords to simulate working ultrasound wands.
3. Place meter stick perpendicular to the edge of the table and the prototype
4. Allow the test subject to reach as far as they can through the hole and record the longest reach they can get to
5. Place plushie on top of the yardstick to simulate the presence of a dragon and record the reach of the subject with the new obstacle
6. Remove the obstacle and test subject from space
7. Measure the distance that the test subject could most reach and place the yard stick horizontally at that distance
8. Have the subject reach into the prototype once more and ask them to reach as far as they can to the sides. Record their reach.
9. Place plushie on top of the yardstick to simulate the presence of a dragon and record the reach of the subject with the new obstacle
10. Remove the obstacle and test subject from space

Test Method 2 - Rating of User Experience

1. Once method one has been completed, ask the subject to rate their experiences on a 1-5 rating scale of 1 being very hard to use and 5 being very easy to use.
2. Ask the test subject to write why they rated their experience as they did and save their responses in element I.
3. Record responses and find the mean of the data points to determine the average opinion of quality.

5. Durability

Test Method 1 - Max shear force on tracks

1. On a given track, vertically attach a spring scale.
2. To the end of the spring scale, attach a 100g mass

3. If the piece is still attached, check the corresponding box for the given total-mass in Table 3.1 to show the piece has passed that test.
4. Repeat steps 2-4 until the total mass on the part is 10kg or the part has separated from the side panel.
5. Repeat steps 1-5 for each track piece and spring latch.

Test Method 2 - Max shear force on springs

1. Pulling down on springs directly
2. Pushing door up while latched in closed position

Test Method 3 - Spring Deformation Test

1. Isolate a spring latch.
2. Measure the equilibrium (unexpanded) length of the spring while vertically oriented. See Figure 3.1 for setup.
3. Attach a 100g mass to the end of the spring such that the full gravitational force on the mass is expanding the spring.
4. Measure and record the expanded length of the spring.
5. Repeat steps 3-4 with masses ranging from 100g to 5kg.

Test Method 4 - Design's reaction to heat/sunlight

1. Measure and record the box's weight, internal temperature, and solidity
2. Place box outside for 1 hour in direct sunlight
3. Record box's final weight, internal temperature, and solidity.

Reliability

Test Method 1 - Spring compression cycling

1. Set door side panel to closed position
- 2.

Test Method 2 - Door drop cycling

Vanderbilt engineer Expert Engineer 3 looked at these testing methods and agreed with the comprehensive testing methods. Expert Engineer 3 further elaborated on these methods and

Population Testing Experiment:

Due to the need for a comprehensive evaluation of many people's experience of the prototype and how it feels to use it, many people need to be surveyed in order to get a full set of data points. The variety of people that we need to survey included in our experiments must be vast in order to reduce confounding variables.